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DeepDILI: Deep Learning-Powered Drug-Induced Liver Injury Prediction Using Model-Level Representation

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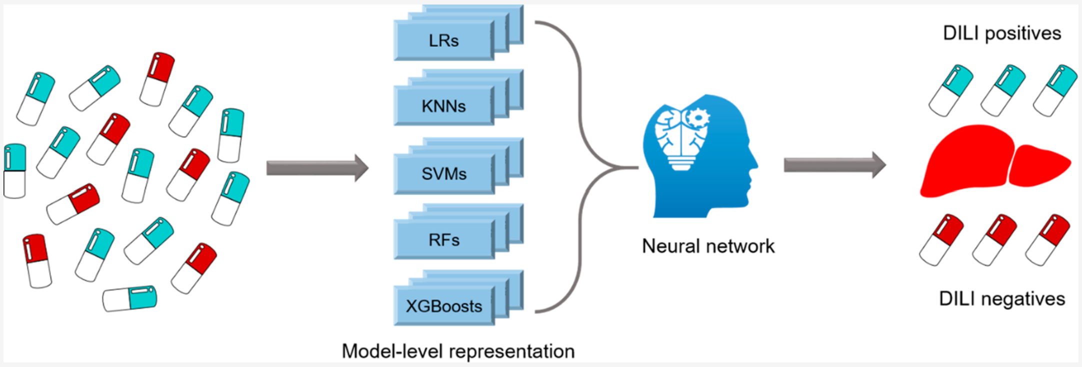

ABSTRACT: Drug-induced liver injury (DILI) is the most frequently reported single cause of safety-related withdrawal of marketed drugs. It is essential to identify drugs with DILI potential at the early stages of drug development. In this study, we describe a deep learning-powered DILI (DeepDILI) prediction model created by combining model-level representation generated by conventional machine learning (ML) algorithms with a deep learning framework based on Mold2 descriptors. We conducted a comprehensive evaluation of the proposed DeepDILI model performance by posing several critical questions: (1) Could the DILI potential of newly approved drugs be predicted by accumulated knowledge of early approved ones? (2) is model-level representation more informative than molecule-based representation for DILI prediction? and (3) could improved model explainability be established? For question 1, we developed the DeepDILI model using drugs approved before 1997 to predict the DILI potential of those approved thereafter. As a result, the DeepDILI model outperformed the five conventional ML algorithms and two state-of-the-art ensemble methods with a Matthews correlation coefficient (MCC) value of 0.331. For question 2, we demonstrated that the DeepDILI model's performance was significantly improved (i.e., a MCC improvement of 25.86% in test set) compared with deep neural networks based on molecule-based representation. For question 3, we found 21 chemical descriptors that were enriched, suggesting a strong association with DILI outcome. Furthermore, we found that the DeepDILI model has more discrimination power to identify the DILI potential of drugs belonging to the World Health Organization therapeutic category of 'alimentary tract and metabolism'. Moreover, the DeepDILI model based on Mold2 descriptors outperformed the ones with Mol2vec and MACCS descriptors. Finally, the DeepDILI model was applied to the recent real-world problem of predicting any DILI concern for potential COVID-19 treatments from repositioning drug candidates. Altogether, this developed DeepDILI model could serve as a promising tool for screening for DILI risk of compounds in the preclinical setting, and the DeepDILI model is publicly available through https://github.com/TingLi2016/DeepDILI.

INTRODUCTION

Drug-induced liver injury (DILI) can be caused by medicines, xenobiotics, and herbs and is one of the leading causes of drug failure in the late stages of clinical trials and withdrawals from the market.[1−3] DILI has been associated with over 1000 drugs and remains the most common reason for acute liver failure in the United States.[4] Several studies have demonstrated that approximately half of the medicines that cause human hepatotoxicity were not identified in preclinical toxicology testing.[5] Global regulatory agencies have advocated a shift toward animal-free testing based on emerging technologies as part of a 21st century risk assessment paradigm. Consequently, alternative approaches for enhancing DILI prediction in the early stage of drug development are being developed.[6] Notable examples include the development of early biomarkers,[7−10] implementation of high-throughput/content assays,[11,12] and

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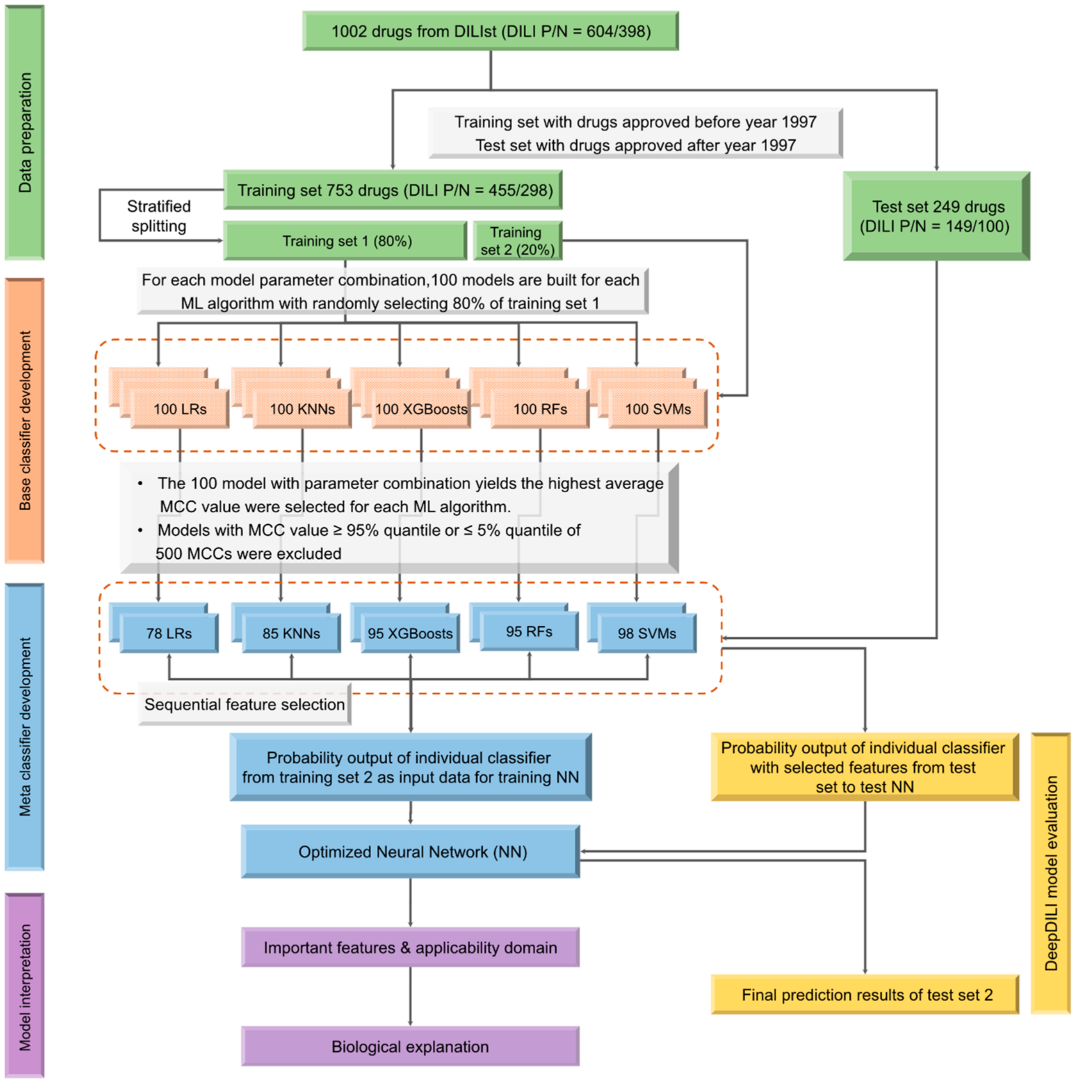

Figure 1. Overall workflow for the DeepDILI model, including: (1) Data preparation: 1002 drugs were split into training (753 drugs) and test (249 drugs) sets based on the initial approval year of drugs with a threshold of 1997. The training set was further split into training sets 1 and 2. (2) Base classifiers development: Base classifiers were developed on the training set 1 through five algorithms, including LR, KNN, SVM, RF, and XGBoost. (3) Meta-classifier development: 451 base classifiers were selected, and the probability prediction of training set 2 was used to train the NN. The sequential feature selection method was applied to find the simplified feature set to develop the DeepDILI model. (4) Model evaluation: The test set further evaluated the DeepDILI model. (5) Model interpretation: Model explainability was established through significant feature enrichment and applicability domain analysis.

utilization of toxicogenomics.[13−15] Although these approaches substantially improved DILI identification in the preclinical setting, synthesis of the drugs and complex experimental testing are still needed. Thus, *in silico* approaches, such as quantitative structure−activity relationship (QSAR) models are increasingly used to detect DILI.

QSAR models have been used extensively in DILI prediction with various descriptors and different machine learning (ML) techniques, including naïve Bayes,[16,17] support vector machine (SVM),[18] decision forest,[19,20] and random forest (RF).[21,22] The performance of these models is not comparable due to the different sizes of data sets, the divergence of the DILI annotations, and the distinct model performance metrics that were employed. Considering the complex underlying mechanism of DILI, ensemble approaches by combining different ML classifiers[23−26] or different DILI-related end points[11,27] were proposed, aimed at maximum capture of DILI-related information to enhance DILI predictions. However, these ensemble strategies are mainly majority voting or average probability based without augmenting combination power from an optimization theory perspective. A few attempts at deep learning models were also made but with limited success.[28] Besides a small sample size, the other prominent reasons for failure could be that the molecule-based representation of deep neural networks (DNNs) could not fully capture the DILI-related information.[29]

For most, if not all, reported QSAR-based DILI prediction models, evaluation is based on the statistical performances which in turn are based on random or stratified data splits accompanied by a strategy involving cross-validation or external validation. This approach does not reflect the real-world scenario where the validity of a model based on past approved drugs has to be verified for its predictivity for future drugs. Moreover, an explainable QSAR model is of great importance for biologists and medical officers to facilitate their decision making. Thus, exploring the features driving the model performance and predefined applicability domains is key to increasing the model explainability and facilitating real-world implementation.

We developed a novel deep learning-powered DILI (DeepDILI) prediction model using the largest binary DILI classification data set (i.e., DILIst)[30] and Mold2 descriptors.[31] The model-level representation generated by five conventional ML algorithms, including *K*-nearest neighbor (KNN), logistic regression (LR), SVM, RF, and extreme gradient boosting (XGBoost) were trained through a neural network (NN) framework for the DeepDILI model development. The DeepDILI model was developed on drugs released to the market pre-1997 and evaluated on the drugs approved thereafter. We comprehensively evaluated the model performance including a comparison with five conventional ML methods, results from standard DNN, and two commonly accepted methods for assemble approaches. We also demonstrated the potential utility of the model in evaluating therapeutic options for COVID-19.

■ MATERIALS AND METHODS

Figure 1 illustrates the overview of the approach taken, which included major five parts: data preparation, model-level representation generation, meta-classifier development, DeepDILI model evaluation, and DeepDILI model interpretation.

Data Preparation. To curate a list of small molecule drugs for DeepDILI model development, we employed DILI severity and toxicity (DILIst), the largest binary DILI annotation data set.[30] First, we curated the DILIst drug information, including drug types, initial approval, and drug identifiers from DrugBank (version 5.1, updated on 2020-01-02),[32] Wikipedia (https://en.wikipedia.org/wiki/List_of_drugs), and the FDALabel database (https://nctr-crs.fda.gov/fdalabel/ui/search). Then, we excluded biologics, mixtures, organometallics, and inorganics from the DILIst. Finally, we downloaded the structure description file of drugs from PubChem (https://pubchem.ncbi.nlm.nih.gov/pc_fetch/pc_fetch.cgi)[33] for molecular descriptor calculation. The final data set consists of 1002 drugs, of which 604 are DILI positives and 398 are DILI negatives (supplementary Table S1). Molecular descriptors of the 1002 drugs were generated using Mold2 (https://www.fda.gov/science-research/bioinformatics-tools/mold2). Mold2 is an open-access software that calculates 777 chemical-physical-based one-/two-dimensional descriptors from chemical structure.[31] We removed the Mold2 descriptors with constant values. Furthermore, we kept only one descriptor if any two descriptors have a pairwise correlation coefficient more than 0.9, while we recorded both descriptor annotations for model interpretation. Consequently, 338 molecular descriptors were selected for model development.

To investigate whether the developed model could utilize the accumulated DILI information from early approved drugs to predict later approved ones, we chronologically divided the 1002 drugs into training and test sets based on the initial approval year. The year 1997 was used as a threshold since the Food and Drug Administration Modernization Act (FDAMA) (https://www.fda.gov/regulatory-information/selected-amendments-fdc-act/food-and-drug-administration-modernization-act-fdama-1997) was implemented at that time. The FDAMA of 1997 aimed to promote regulatory evaluation by adopting emerging technologies and eliminate unexpected adverse drug reactions in drug products. The drugs approved before and after 1997 were divided into a training set and test set, respectively. Accordingly, the training set consisted of 753 drugs (455 DILI positive/298 DILI negative), and the test set included 249 drugs (149 DILI positive/100 DILI negative).

DeepDILI Model Development. DeepDILI is an ensemble classifier with a NN (Figure 1). The probability outputs from base classifiers (model-level representation) are the input for the second level of NN. This approach tries to use the meta-classifier to learn the complementary information provided by the base classifiers,[23,34,35] based on the hypothesis that no single learning algorithm is optimum at any modeling circumstances. Thus, the model performance of ensemble classifiers can be improved to some extent.

Base Classifier Development. Five algorithms, including LR, KNN, SVM, RF, and XGBoost, were used for base classifier development. The brief introduction of the five ML algorithms are provided in Table 1. To develop the base classifiers, we first randomly stratified split the training set (753 drugs) into training set 1 (753 × 80% = 602 drugs) and training set 2 (753 × 20% = 151 drugs). Then, for each algorithm, we comprehensively optimized the hyperparameters by using a bootstrap aggregating strategy.[36] Specifically, for each model and hyperparameter combination, 100 models were developed by randomly selecting 80% of drugs in training set 1 and then predicting the drugs in training set 2. For each algorithm, the 100 models with hyperparameter combination yielding the highest average Matthews correlation coefficient (MCC) were selected. Next, we ranked order the 500 base classifiers (100 models × 5 algorithms = 500 models) from high to low based on their MCC values. To avoid overfitting or underfitting problems of the developed base classifiers, we only kept the models with MCC values in the range of 5−95% percentile of the 500 models as the optimized base classifiers.

Meta-Classifier Development. The NN aims to find the correct mathematical expression to transform the input into the output through a linear or a nonlinear relationship. The network moves through the layers by calculating the probability of each output. In this study, we developed a three-layer NN model as a meta-classifier for DILI prediction. Specifically, we extracted the probability output of the optimized base classifiers (i.e., model-level representation) from the training set 2 as the NN input layer. In other words, the model-level representation was constructed by a vector consisting of the predicted probability values from the filtered based classifiers. Following a hidden layer with 10 nodes with hyperbolic tangent activation (tanh) activation, and Adam optimizer, the output layer is used to project the hidden layer information into probabilistic values of DILI prediction using sigmoid function. The learning rate is set to 0.0001. The proposed DeepDILI model is a stacked model by combining optimized-based classifiers and NN-based meta-classifiers. For a new test compound, the compound's model-level representation is generated by the optimized base classifiers, serving as the input of the trained meta-classifier to generate prediction value.

DeepDILI Model Evaluation. *Independent Test Set.* We used 249 drugs (149 DILI positive/100 DILI negative) in the test set to investigate the model performance of the developed DeepDILI model. We employed the area under the receiver operating characteristic (ROC) curve analysis, which exhibits the performance of the classification model with a plot by the true predictive rate (TPR) against the false positive rate (FDR).[37] We calculated the area under the ROC curve (AUC) for each developed model. We also used six other performance metrics including accuracy, sensitivity, specificity, F1, MCC, and balanced accuracy (BA) for further evaluation of model performance, which were computed using eqs 1−6:

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FN} + \text{FP}} \tag{1}$$

$$\text{sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

$$\text{specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{3}$$

$$\text{F1} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{4}$$

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP}) \times (\text{TP} + \text{FN}) \times (\text{TN} + \text{FP}) \times (\text{TN} + \text{FN})}} \tag{5}$$

$$\text{BA} = \frac{\text{sensitivity} + \text{specificity}}{2} \tag{6}$$

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Table 1. Descriptions of Five ML Algorithms Used for Base Classifiers Development

method	description
logistic regression (LR)	LR is a statistical model that uses a logistical function of a linear combination of multiple independent variables to model a binary-dependent variable.[55] The output probability indicates the chance of an object belonging to a certain class. In this study, drugs predicted with probability ≥0.5 belong to DILI positives, and the rest belongs to DILI negatives. The inverse regularization parameter (C) in LR was optimized.
K-nearest neighbor (KNN)	KNN is an instance-based learning algorithm that classifies test objects based on the closest objects from the training set.[56] The class label of the test object is determined by the majority of the class label of its K-nearest neighboring objects. In this study, the Euclidean distance was applied to measure the distance between the object and its neighbor. The hyperparameter (i.e., the number of neighbors K) was optimized.
support vector machine (SVM)	SVM is a learning algorithm based on the Vapnik−Chervonenkis (VC) theory.[57] It constructs a hyperplane in a high-dimensional space for classification purposes. The hyperplane separates the two classes but maintains the maximal distance (margin) from any object. SVM can perform nonlinear classification by applying kernel functions. The kernel function projects the object into higher-dimensional space, which can help to separate classes. The radial basis function (RBF) was used as the kernel function in our study. The polynomial and RBF kernel functions with hyperparameters, i.e., γ for the nonlinear kernel and penalty parameter C, were optimized.
random forest (RF)	RF is an ensemble algorithm for classification.[58] It constructs a large number of decision trees. A decision tree is built by bootstrapping training objects and randomly selecting subsets of original independent variables. The tree-building method makes the diversity of the forest. Every decision tree classifies the new object, and the final classification result is decided by the majority voting of all the decision trees. We optimized the hyperparameters, including the number of trees, the max depth of the tree, the minimum sample split, and the minimum sample leaf.
extreme gradient boosting (XGBoost)	XGBoost is a special implementation of gradient boosted trees algorithm.[59] A new tree that predicts the errors or residuals of prior trees are combined with the previous trees sequentially until no further improvements can be made. The gradient descent algorithm is applied to minimize the loss when adding new trees. Compared with the original gradient boosted trees algorithm, XGBoost has four unique attributes (highly scalable end-to-end tree boosting system; weighted quantile sketch; sparsity-aware algorithm, effective cache-aware block structure) for model construction. Hyperparameters, including learning rate, estimators, max depth, and subsample, were optimized.

where TP, TN, FP, and FN denote true positive, true negative, false positive, and false negative, respectively.

Comparative Analysis with Other Modeling Approaches. Four comparative studies were conducted to assess the DeepDILI method. First, we compared DeepDILI against base classifiers. We developed base models for the five conventional ML algorithms (i.e., LR, KNN, SVM, RF, and XGBoost) with the drugs in the whole training set, and then the optimized model was used to predict the drugs in the test set.

Second, we compared DeepDILI against the molecule representation-based DNN models. We used the whole training set to train the conventional DNN model with different hyperparameter combinations among optimizers, activations, and learning rates. Then, the optimized DNN model was employed to predict the drugs in the test set.

Third, we compared DeepDILI against a published DNN model using the molecule representation for DILI predictions.[28] In this reported study, three independent different DILI data sets, including NCTR LTKB benchmark data set (95 DILI positive and 90 DILI negative),[38] Greene's data set (209 DILI positive and 111 DILI negative),[39] and Xu's data set (128 DILI positive and 108 DILI negative)[40] were used. We excluded the compounds in three data sets overlapping with the training set to generate three independent external data sets. Consequently, we obtained 57 compounds (i.e., 35 DILI positives and 22 DILI negatives), 80 compounds (i.e., 52 DILI positives and 28 DILI negatives), and 64 compounds (34 DILI positives and 30 DILI negatives) for the NCTR data set, the Greene's data set, and the Xu's data set, respectively (Supplementary Table S1). We evaluated the DeepDILI model on these three data sets and compared the prediction performance with the reported results.

Fourth, we compared DeepDILI against other ensemble approaches including both the majority voting method and average probability method. For the majority voting method, the classification label for a new object is decided by the majority class of the predictions made by all the base classifiers. For the average probability method, the probability prediction of a new object from all the base classifiers was collected and averaged. An object with the average probability <0.5 belongs to DILI negative and otherwise belongs to DILI positive.

DeepDILI Interpretability. To obtain an interpretable model, we employed a sequential feature selection approach consisting of two steps:

(1) Rank order the individual descriptor based on its model contribution. Specifically, we removed one descriptor from the total 338 chemical descriptors in the optimized base classifiers at once and generated model-level representation in training set 2. The model-level representation was used as an input to train the three-layer NN model described above. A higher area under the curve (AUC) indicates less influence on the model performance by the removed feature. We further computed model contribution (i.e., the percentage of AUC changes) by comparing the AUC generated from the removed descriptor-based and all the 338 chemical descriptors-based NN models, as shown in eq 7:

$$\text{model contribution}_i = \frac{\text{AUC}_w - \text{AUC}_i}{\text{AUC}_w} \quad (7)$$

where AUC_i denotes the AUC value of the NN model developed without descriptor i, and AUC_w represents the AUC value of the NN model developed by all the 338 descriptors.

(2) Stepwise feature selection. First, we ranked order the descriptors based on the model contribution from low to high. Then, we divided the ranked 338 descriptors into 23 bins with approximately 15 descriptors per bin. We sequentially removed the bin of descriptors and developed the NN model with the rest of the descriptors to predict drugs in training set 2. The objective is to remove as many as uninformative descriptors with minimum loss of prediction performance. Accordingly, we developed an optimized feature score (OFS)

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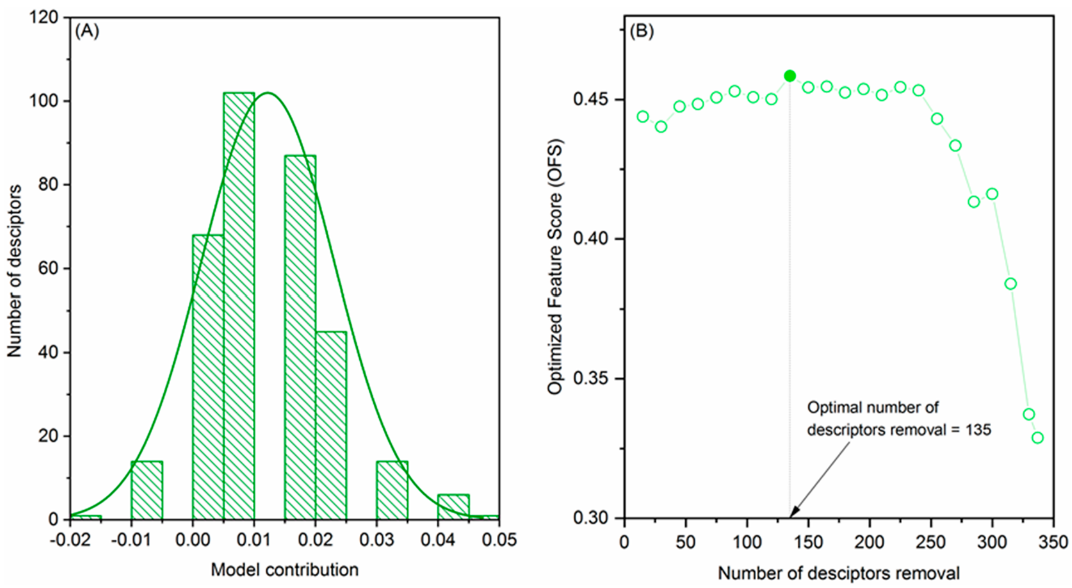

Figure 2. (A) the distribution of model contribution of the 338 Mold2 descriptors. (B) The relationship between the number of descriptor removal and OFS.

by combining the model performance (i.e., AUC) and number of removed features, as shown in the eq 8:

$$\text{optimized feature score} = \frac{\mathrm{AUC}_n - c \times \left(\frac{n}{N}\right)}{2} \tag{8}$$

where n denotes the number of descriptors removal, and N is the total number of descriptors (i.e., 338 descriptors). The parameter c is penalty constant, and here we set it to 0.15. AUC_n represents the model performance of the NN models with n descriptors removal from base classifiers. The higher the OFS is, the more preservability of the model performance is kept with features removed.

Impact of Molecular Descriptors on DeepDILI Model Performance. To investigate the impact of different molecular descriptors on DeepDILI model performance, we employed two additional molecular descriptors (i.e., Mol2vec[41] and Molecular ACCess System (MACCS) structural keys[42]). Mol2vec is 300 bits long vector representations of molecular substructure using unsupervised pretraining methods to obtain high-dimensional embeddings. MACCS is a 166 bits binary fingerprint to indicate the presence of 166 predefined substructures encoded as SMARTS patterns. We followed the same model development procedure for DeepDILI based on Mold2 descriptors to develop the DeepDILI model based on Mol2vec and MACCS. Then, we compared the DeepDILI models developed by the three different types of molecular descriptors with the feature-based six-layer DNN model and published DNN model on the test set and three external test sets, respectively.

DeepDILI Applicability Domain. To define the applicability domain of the proposed DeepDILI model, we calculated the model performance of drugs based on their therapeutic classes in the test set. Specifically, the World Health Organization (WHO) Anatomical Therapeutic Chemical (ATC) classification system was used to define therapeutic drug class.[43] The WHO-ATC classification system is a five-level hierarchical ontology that categorized drugs based on the drugs' therapeutic, pharmacological, and chemical properties. In this study, we mapped the drugs in the test set into the first ATC level, including 14 main anatomical/pharmacological groups. Only the therapeutic classes with drugs number more than 20 were kept for the analysis leaving six therapeutic classes including: A, alimentary tract and metabolism; J, anti-infectives for systemic use; C, cardiovascular system; L, anti-neoplastic and immunomodulating agents; G, genito-urinary system and sex hormones; and N, nervous system.

DeepDILI for DILI Assessment of Reported Repurposing Candidates for COVID-19. The ongoing COVID-19 pandemic poses a global health crisis that could be addressed by repositioning existing drugs for COVID-19 treatment, potentially with a quicker drug development turnaround.[44] DILI has been associated with COVID-19 patients where elevated lactate dehydrogenase, aspartate aminotransferase, and alanine aminotransferase[45] have been reported. Therefore, it is critical to screen repurposed candidates for DILI potential to ensure safer treatment development. We used the developed DeepDILI model to predict the DILI potential for repurposed candidates curated by DrugBank.[32] There was a total of 39 repurposing candidates for COVID-19 in the Drugbank COVID-19 information (https://www.drugbank.ca/covid-19#drugs). Once we excluded vaccines, monoclonal and compounds in our training, there were 16 small molecule COVID-19 repurposing candidates in the assessment.

DeepDILI for Screening DILI potential of Compounds in the DrugBank Database. The developed DeepDILI model was further utilized to screen the DILI potential of compounds in the Drugbank database.[46] Specifically, a total of 10,767 approved drugs and investigational compounds were downloaded through https://go.drugbank.com/releases/latest#structures. We excluded the compounds that are overlapped with our training set. Consequently, 10,026 compounds were obtained to be screened by the proposed DeepDILI model (Supplementary Table S1). We used the predicted probabilistic values yielded from the DeepDILI model to quantitatively measure the DILI concerns. Considering the lack of ground truth of DILI concerns for the investigational compounds, we arbitrarily defined the DILI concerns into five categories: high DILI concerns (i.e., predicted probabilistic values >0.95), likely high DILI concerns (i.e., predicted probabilistic values in a range: 0.75−0.95), uncertain (i.e., predicted probabilistic values in a range: 0.35−0.75), likely less-DILI (i.e., predicted probabilistic values in a range: 0.15−0.35), and no-DILI concern (i.e., predicted probabilistic values <0.15).

Code Availability. The open-source Python (version 3.6.5) was used to develop all the models introduced above. The source code of the Mol2vec descriptors was downloaded from https://github.com/samoturk/mol2vec. The MACCS fingerprints were generated from RDKit (version: 2020.09.1) in Python, an open-source cheminformatics toolkit. The NN classifier was developed with the Keras library version 2.0 with TensorFlow version 1.14 as the backend. The models based on algorithms of KNN, LR, SVM, and RF were developed using the scikit-learn package version 0.22.[47] Models of XGBoost were developed using its open-source library implemented on Python (version 3.6.5). The model scripts of this study are available at https://github.com/TingLi2016/DeepDILI.

■ RESULTS

Base Classifier Development. To obtain the optimized base classifiers for model-level representation generation, we comprehensively evaluated model performance under different hyperparameter combinations (Supplementary Table S2). For each ML classifier with a specific hyperparameter combination, 100 models were generated by randomly selecting 80% of drugs in the training set 1 and predicting the drugs in training set 2. The 100 models with hyperparameter combination yielding the highest average MCC values are selected for further analysis. For LR, the optimized C was 0.1. For KNN, the optimized number of neighbors K was 7. For SVM, the optimized hyperparameter combination of kernel, C, and γ was the radial basis function (RBF) kernel, 1, and scale. For RF, the best hyperparameter combination of *estimators*, *max depth*, and *min_samples_leaf* was 700, 11, and 5, respectively. For XGBoost, the best hyperparameter combination of the learning rate, estimators, max depth, and subsample was 0.01, 700, 11, and 0.7. We further excluded 500 models with optimized hyperparameter combinations (100 model × 5 ML classifiers = 500 models) with a MCC >95% percentile (0.351) and <5% percentile (0.128) to avoid the model suffering over/under-fitting problems. Consequently, there are a total of 451 models, including 78 LR models, 85 KNN models, 98 SVM models, 95 RF models, and 95 XGBoost models, that are considered as final base classifiers. The complex algorithms (e.g., SVM, RF, and XGBoost) have more models selected as the base classifiers than the simple ML algorithms such as LR and KNN, indicating the complex models may be more robust for this data set.

DeepDILI Model Development. DeepDILI model is an ensemble model that generates a model-level representation from base classifiers using an NN in DILI prediction. Specifically, the probabilistic values of drugs from base classifiers serve as the NN model's input. To increase model explainability, we carried out a sequential feature selection to exclude the uninformative descriptors from the base classifiers (see Materials and Methods section). Figure 2A illustrates the distribution of individual descriptor contribution to the DeepDILI model. There were 21 Mold2 descriptors that individually contributed more than 3% to the model (Table 2 and Supplementary Table S3). Most of these 21 descriptors were functional groups and topological structure related, such as '*number of group secondary amides (aromatic)*'.

Figure 2B depicts the relationship between the number of descriptors removal and the OFS. With ranked individual descriptors based on model contribution from high to low, we developed the NN model by sequentially removing the ranked descriptors from base classifiers, and the OFS was calculated accordingly. The larger OFS indicated the NN model performance could be preserved with features removed. The OFS reached the maximum (0.458) when 135 descriptors were removed. As a result, we obtained the optimized DeepDILI model with 203 (full set descriptors: 338; removed descriptors: 135) descriptors for base classifiers and a trained NN as a meta-classifier.

The model performance of the developed DeepDILI model and base classifiers on the training set 2 is listed in Table 3. For each ML algorithm, the performance results were averaged,

Table 2. Descriptors with a Model Contribution of more than 3% for the Mold2-DeepDILI

Mold2 descriptor ID	model contribution (%) (range: −1.60% to 4.91%)	description
D626	4.91	number of group secondary amides (aromatic)
D776	4.17	aromatic bonds ratio
D498	4.17	Moran topological structure autocorrelation length-4 weighted by atomic Sanderson electronegativities
D510	4.17	Moran topological structure autocorrelation length-8 weighted by atomic polarizabilities
D737	4.17	number of group R−CH−R
D680	4.17	number of group ethers (aromatic)
D619	4.17	number of group carboxylic acids (aliphatic)
D750	3.31	number of group H attached to C0(sp^3) with 3X attached to next C
D748	3.31	number of group H attached to C0(sp^3) with 1X attached to next C
D739	3.31	number of group R−CX−R
D003	3.31	number of four-membered rings
D530	3.31	mean molecular topological order-10 charge index
D662	3.31	number of group ammonia groups (aromatic)
D676	3.31	number of group primary alcohols (aliphatic)
D545	3.31	lowest eigenvalue from Burden matrix weighted by van der Waals order-6
D596	3.31	number of total primary C-sp^3
D488	3.31	Moran topological structure autocorrelation length-2 weighted by atomic van der Waals volumes
D015	3.31	rotatable bond fraction
D474	3.31	Geary topological structure autocorrelation length-4 weighted by atomic polarizabilities
D528	3.31	mean molecular topological order-8 charge index
D575	3.31	highest eigenvalue from Burden matrix weighted by van der Waals order-4

and the standard deviation of the optimized base classifiers with selected 203 descriptors was calculated. DeepDILI model outperformed all of the individual base classifiers across almost all the performance metrics. For example, the DeepDILI has approximately 88−179% improvement of the MCC value compared to the base classifiers. The MCC values of base classifiers were ranked in the following order: SVM (0.286 ± 0.042) > LR (0.285 ± 0.036) > RF (0.211 ± 0.044) > XGBoost (0.209 ± 0.038) > KNN (0.193 ± 0.054). The sensitivity of DeepDILI (0.846) was lower than that of LR and SVM (0.930 ± 0.020 and 0.888 ± 0.034). However, LR and SVM yielded a very unbalanced sensitivity (0.930 ± 0.020 and 0.888 ± 0.034) and specificity (0.280 ± 0.036 and 0.346 ± 0.037).

The DeepDILI Model Could Predict DILI Potential of Drugs Approved in the Later Years. To further investigate whether the DeepDILI approach could generate a model based on drugs approved in early years to predict these approved in the later years, we developed a DeepDILI model based on pre-1997 and evaluated on the drugs (i.e., the test set) that were approved thereafter (Table 3). DeepDILI yielded the highest MCC (0.331), accuracy (0.687), AUC (0.659), F1 score (0.755), balanced accuracy (0.658), and specificity (0.510),

Table 3. Model Performances of the DeepDILI Model and Base Classifiers on Training Set 2 and Test Set

models	MCC		accuracy		AUC		F1 score		BA		sensitivity		specificity	
	training set 2	test set	training set 2	test set	training set 2	test set	training set 2	test set	training set 2	test set	training set 2	test set	training set 2	test set
Mold2 Descriptors														
LR	0.285 ± 0.036	0.130 ± 0.038	0.671 ± 0.013	0.617 ± 0.011	0.582 ± 0.015	0.628 ± 0.009	0.773 ± 0.009	0.744 ± 0.007	0.605 ± 0.015	0.54 ± 0.013	0.930 ± 0.020	0.932 ± 0.017	0.280 ± 0.036	0.147 ± 0.033
KNN	0.193 ± 0.054	0.125 ± 0.038	0.62 ± 0.026	0.582 ± 0.019	0.614 ± 0.021	0.580 ± 0.021	0.695 ± 0.024	0.657 ± 0.020	0.594 ± 0.027	0.562 ± 0.019	0.718 ± 0.036	0.669 ± 0.035	0.470 ± 0.044	0.454 ± 0.038
SVM	0.286 ± 0.042	0.220 ± 0.035	0.673 ± 0.017	0.646 ± 0.013	0.569 ± 0.015	0.645 ± 0.009	0.766 ± 0.015	0.752 ± 0.008	0.617 ± 0.016	0.584 ± 0.017	0.888 ± 0.034	0.899 ± 0.026	0.346 ± 0.037	0.268 ± 0.052
RF	0.211 ± 0.044	0.225 ± 0.030	0.636 ± 0.02	0.645 ± 0.013	0.617 ± 0.018	0.658 ± 0.012	0.721 ± 0.018	0.736 ± 0.009	0.598 ± 0.020	0.600 ± 0.014	0.782 ± 0.032	0.828 ± 0.013	0.414 ± 0.033	0.371 ± 0.028
XGBoost	0.209 ± 0.038	0.219 ± 0.037	0.635 ± 0.017	0.642 ± 0.016	0.640 ± 0.017	0.651 ± 0.015	0.719 ± 0.015	0.732 ± 0.012	0.598 ± 0.018	0.598 ± 0.018	0.776 ± 0.026	0.818 ± 0.021	0.420 ± 0.037	0.378 ± 0.036
DeepDILI	0.538	0.331	0.781	0.687	0.857	0.659	0.824	0.755	0.765	0.658	0.846	0.805	0.683	0.510
conventional DNN	–	0.263	–	0.659	–	0.653	–	0.740	–	0.621	–	0.812	–	0.430
Mol2vec Descriptors														
KNN	0.185 ± 0.043	0.170 ± 0.035	0.615 ± 0.02	0.612 ± 0.016	0.608 ± 0.019	0.605 ± 0.017	0.689 ± 0.018	0.693 ± 0.014	0.591 ± 0.021	0.582 ± 0.017	0.708 ± 0.029	0.734 ± 0.023	0.474 ± 0.041	0.429 ± 0.032
LR	0.182 ± 0.038	0.186 ± 0.051	0.600 ± 0.021	0.617 ± 0.023	0.620 ± 0.019	0.624 ± 0.014	0.654 ± 0.024	0.694 ± 0.018	0.592 ± 0.019	0.590 ± 0.025	0.628 ± 0.035	0.726 ± 0.025	0.556 ± 0.029	0.454 ± 0.044
RF	0.222 ± 0.037	0.200 ± 0.029	0.640 ± 0.017	0.631 ± 0.012	0.624 ± 0.017	0.661 ± 0.013	0.722 ± 0.014	0.719 ± 0.009	0.604 ± 0.017	0.592 ± 0.014	0.776 ± 0.024	0.791 ± 0.016	0.432 ± 0.032	0.393 ± 0.032
SVM	0.189 ± 0.044	0.172 ± 0.038	0.637 ± 0.015	0.628 ± 0.012	0.614 ± 0.022	0.639 ± 0.017	0.752 ± 0.011	0.752 ± 0.009	0.566 ± 0.022	0.551 ± 0.016	0.912 ± 0.042	0.944 ± 0.031	0.220 ± 0.076	0.157 ± 0.054
XGBoost	0.189 ± 0.044	0.175 ± 0.041	0.627 ± 0.020	0.624 ± 0.017	0.639 ± 0.019	0.64 ± 0.016	0.716 ± 0.017	0.721 ± 0.012	0.587 ± 0.02	0.577 ± 0.019	0.780 ± 0.028	0.813 ± 0.022	0.395 ± 0.036	0.341 ± 0.040
DeepDILI	0.413	0.293	0.722	0.671	0.786	0.654	0.774	0.745	0.704	0.638	0.791	0.805	0.617	0.470
conventional DNN	–	0.289	–	0.663	–	0.685	–	0.725	–	0.642	–	0.745	–	0.540
MACCS Descriptors														
KNN	0.217 ± 0.037	0.110 ± 0.037	0.641 ± 0.016	0.589 ± 0.017	0.624 ± 0.017	0.586 ± 0.015	0.729 ± 0.014	0.685 ± 0.015	0.599 ± 0.017	0.551 ± 0.017	0.804 ± 0.025	0.747 ± 0.027	0.394 ± 0.033	0.355 ± 0.032
LR	0.260 ± 0.031	0.198 ± 0.028	0.633 ± 0.014	0.606 ± 0.014	0.665 ± 0.009	0.636 ± 0.007	0.677 ± 0.013	0.657 ± 0.015	0.633 ± 0.016	0.601 ± 0.014	0.637 ± 0.018	0.630 ± 0.024	0.628 ± 0.027	0.571 ± 0.032
RF	0.265 ± 0.032	0.177 ± 0.036	0.646 ± 0.016	0.611 ± 0.017	0.669 ± 0.009	0.643 ± 0.009	0.704 ± 0.015	0.686 ± 0.015	0.633 ± 0.016	0.587 ± 0.018	0.697 ± 0.025	0.710 ± 0.027	0.569 ± 0.031	0.463 ± 0.039
SVM	0.236 ± 0.037	0.098 ± 0.039	0.652 ± 0.016	0.601 ± 0.013	0.625 ± 0.014	0.611 ± 0.015	0.746 ± 0.014	0.721 ± 0.009	0.602 ± 0.016	0.538 ± 0.017	0.847 ± 0.031	0.859 ± 0.019	0.357 ± 0.039	0.217 ± 0.041
XGBoost	0.219 ± 0.036	0.120 ± 0.036	0.625 ± 0.018	0.589 ± 0.017	0.647 ± 0.017	0.616 ± 0.014	0.687 ± 0.017	0.678 ± 0.014	0.61 ± 0.018	0.557 ± 0.018	0.682 ± 0.027	0.722 ± 0.023	0.537 ± 0.032	0.393 ± 0.032
DeepDILI	0.354	0.267	0.695	0.647	0.737	0.640	0.755	0.703	0.673	0.634	0.780	0.698	0.567	0.570
conventional DNN	–	0.213	–	0.618	–	0.605	–	0.676	–	0.607	–	0.664	–	0.550

with an improvement range of around 47−165% for MCC. Unlike the rank order of ML algorithms base on the training set 2, we found that the ranking order on MCC values of base classifiers on the test set in a sequence of RF (0.225 ± 0.030) > SVM (0.220 ± 0.035) > XGBoost (0.219 ± 0.037) > LR (0.130 ± 0.038) > KNN (0.125 ± 0.038). Notably, LR and KNN's performances were dramatically decreased in the test set compared to training set 2, suggesting the complex algorithms (i.e., RF, XGBoost, SVM) may have better model generality than simple ML classifiers (i.e., LR and KNN). To further confirm the finding, we retrained the five conventional ML classifiers with the whole training set and then predicted the test set. The performances of complex algorithms (i.e., RF, XGBoost, SVM) showed no significant divergence between the training and test sets. However, the performances of simple ML classifiers (i.e., LR and KNN) were not well preserved between the two sets (Supplementary Table S4). Furthermore, DeepDILI provided a more balanced sensitivity of 0.805 and specificity of 0.510 than any other base classifier. Collectively, we demonstrated the DeepDILI model could combine the information from base classifiers and yielded a better performance for predicting newly approved drugs with accumulative DILI information.

DeepDILI with Model-Level Representation Outperformed Molecule Representation-Based DNN. To investigate the advantage of model-level presentation over feature-level representation in DILI prediction, we compared the DeepDILI model with conventional DNNs (Table 3). First, we developed a six-layer DNN model with the whole training set with optimized hyperparameters (i.e., learning rate: 0.0001; optimizer: Adam; activation function: tanh for input and hidden layers and sigmoid for output layer; batch size = 8; and dropoff rate: 0.2) to predict drugs in the test set. The DeepDILI model (MCC = 0.331) outperformed the DNN model (MCC = 0.263) with an improvement of 25.86%, indicating the model-level representation capture more DILI-related information. Furthermore, we applied the DeepDILI model to three independent DILI data sets and compared the results with the published DNN model with molecule-level representation[28] (Figure 3). Three performance metrics including accuracy, sensitivity, and specificity were used in the published DNN model. The accuracy achieved by the

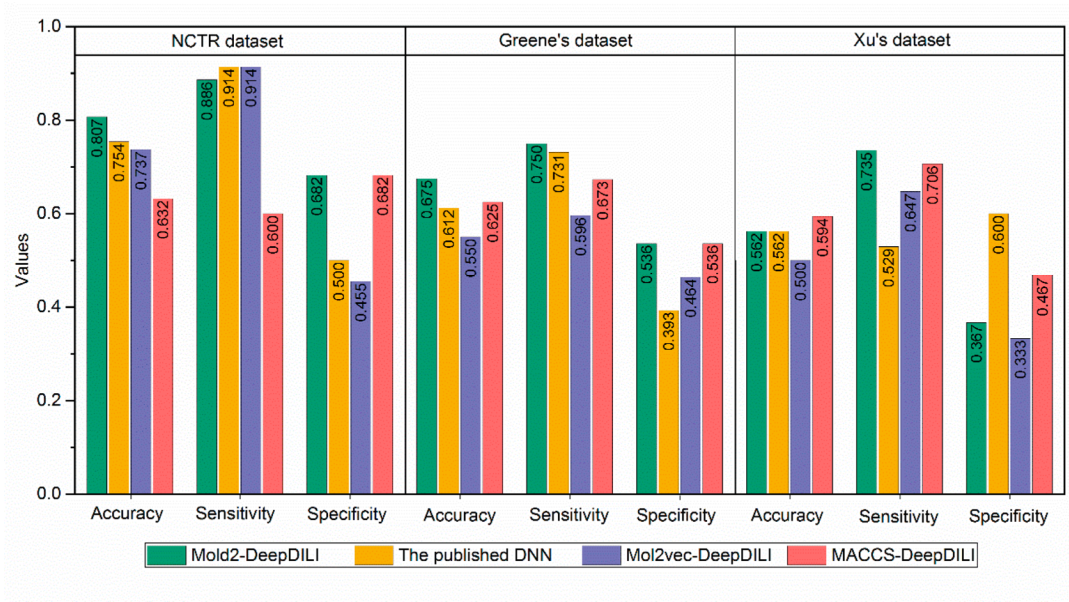

Figure 3. Comparison of DeepDILI model with the published DNN. For three independent data sets including NCTR data set, Greene's data set, and Xu's data set, the published DNN model (yellow) was employed. DeepDILI models were developed by three molecular descriptors, including Mold2 (green), Mol2vec (purple), and MACCS (pink).

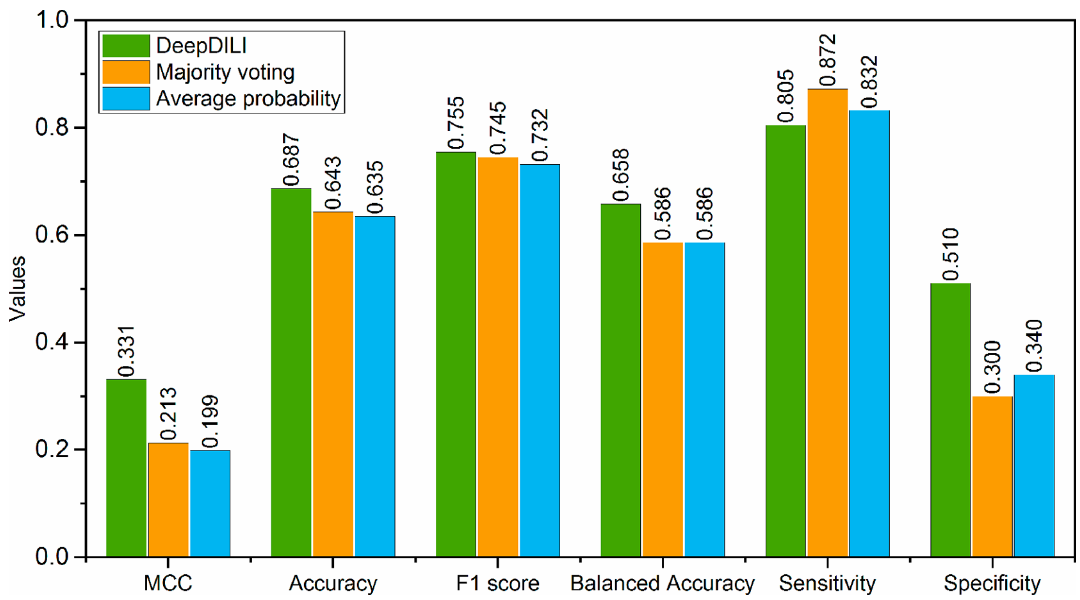

Figure 4. Ensemble models performance on the test set. Seven performance metrics for DeepDILI model (green color), the majority voting model (yellow color), and the average probability model (blue color)

DeepDILI model was significantly higher than by the published DNN model on the NCTR and Greene's data set and comparable performance on Xu's data set. Specifically, the DeepDILI model yielded a respective accuracy of 0.807 and 0.675 for the NCTR data set and Greene's data set, which improved 7.0% and 10.3% compared to the published DNN model (Figure 3).

DeepDILI Is Significantly Better than State-of-the Art Ensemble Methods. We also conducted a comparative study between the DeepDILI model and two state-of-the-art ensemble approaches on the test set, that is, majority voting and average probability (Figure 4). Except for sensitivity, the DeepDILI model outperformed the two ensemble approaches across all of the other six performance metrics with an improvement of 55.40%, 6.84%, 1.34%, 12.29%, and 50.00% on MCC, accuracy, F1, BA, and specificity, respectively. Moreover, the majority voting and average probability strategies provided unbalanced sensitivity (0.872 and 0.832) and specificity (0.300 and 0.340).

DeepDILI Using Mold2 Descriptors Outperformed DeepDILI Using Mol2vec and MACCS Descriptors. Table 3 lists the prediction performance of DeepDILI models developed by the three molecular descriptors on the test set. The DeepDILI models outperformed the base classifiers for all three types of descriptors. The prediction performance of DeepDILI models on the test set was ranked in the following order: Mold2-DeepDILI > Mol2vec-DeepDILI > MACCS-DeepDILI across all the performance metrics except specificity.

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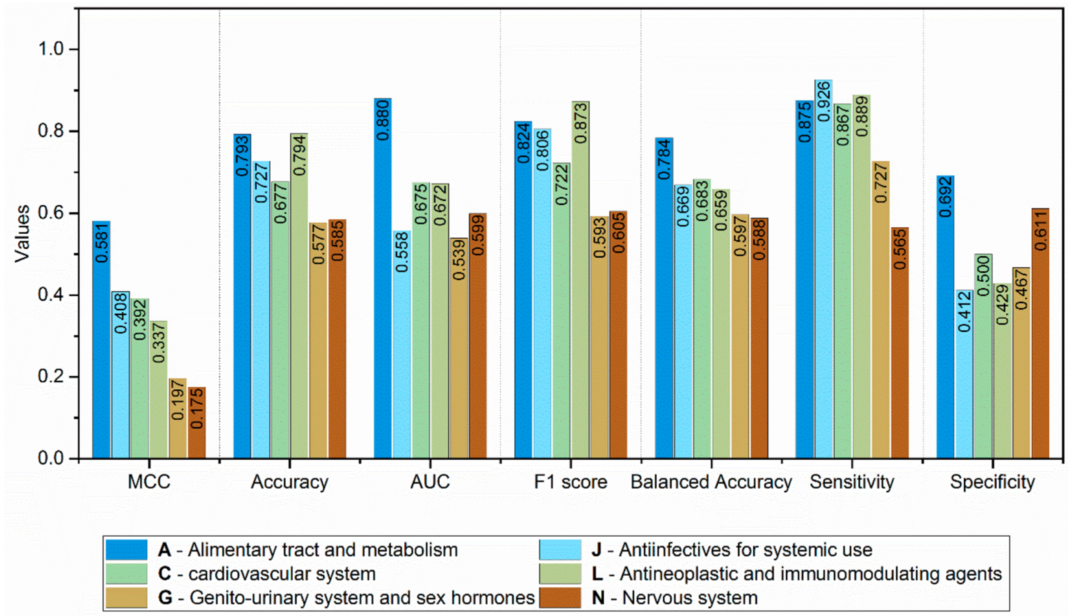


Figure 5. DeepDILI model performance of drugs from the six therapeutic classes according to the first level of the WHO ATC codes: A, alimentary tract and metabolism; J, anti-infectives for systemic use; C, cardiovascular system; L, anti-neoplastic and immunomodulating agents; G, Genito-urinary system and sex hormones; and N, nervous system.

For example, The Mold2-DeepDILI model achieved the highest MCC of 0.331, followed by the Mol2vec-DeepDILI model and MACCS-DeepDILI model with the MCC of 0.293 and 0.267, respectively. The MACCS-DeepDILI model yielded the highest specificity of 0.570 over 0.510 and 0.470 of the Mold2-DeepDILI and Mol2vec-DeepDILI. Furthermore, all three DeepDILI models (i.e., Mold2-DeepDILI, Mol2vec-DeepDILI, and MACCS-DeepDILI) outperformed the six-layer DNN model with improvements of 25.86%, 1.38%, and 25.35% in MCC, respectively. Also, the ranked features of Mol2vec and MACCS were ranked based on their model contribution were listed in the Supplementary Table S3.

Figure 3 illustrates the prediction performance of DeepDILI models and the published DNN model[28] on the three external test sets. The mold2-DeepDILI achieved the highest accuracies (0.807 and 0.675) on the NCTR and Greene's data sets, improving 7.0% and 10.3% over the published DNN model. The MACCS-DeepDILI model yielded the highest accuracy (0.594) on Xu's data set, and mold2-DeepDILI and the published DNN model obtained the same accuracy (0.562). Furthermore, the Mold2-DeepDILI tended to generate a more balanced sensitivity and specificity based on the observation from NCTR and Greene's data sets.

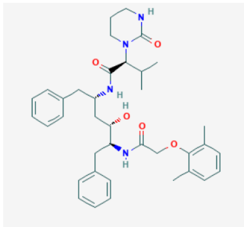
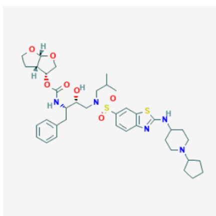
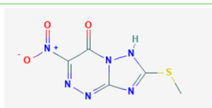
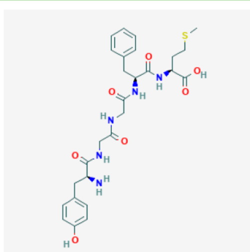
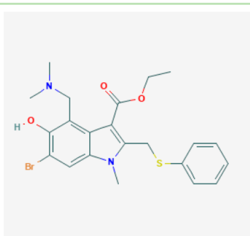
The Mold2-DeepDILI model provided superior performance to DeepDILI models developed by the other two molecular descriptors. Therefore, we used the Mold2-DeepDILI model for further analyses of the applicability domain and real-world applications.

DeepDILI Model Yielded a Better Performance for Alimentary Tract and Metabolism Drugs. We evaluated the performance of DeepDILI across various therapeutic applications to assess its applicability domain based on the test set. Figure 5 shows the prediction performance of the six WHO-ATC therapeutic classes, including A, J, C, L, G, and N. The predictive performance of the DeepDILI model varied in each class. The category A outperformed other therapeutic classes, with a higher MCC, accuracy, AUC, F1, BA, sensitivity, and specificity of 0.581, 0.793, 0.880, 0.824, 0.784, 0.875, and 0.692, respectively. Compared with the DeepDILI model performance on the test set, the category A's performance metrics improved 75.53%, 15.43%, 33.54%, 9.14%, 19.15%, 8.70%, and 35.69%, respectively. However, the lower predictive performances (MCC < 0.2) were observed in some anatomical/pharmacological categories, such as G and N.

DeepDILI Model Could Be Utilized to Address DILI Concerns for COVID-19 Drug Repositioning. Table 4 lists the prediction results of the 16 COVID-19 repurposing candidates using the DeepDILI model. Of the 16 COVID-19 repurposing candidates, there were 7 drugs with predicted high probabilistic values between 0.75 and 0.95, 6 drugs with predicted high probabilistic values in the range of 0.35−0.75, 1 drug with predicted probabilistic values in the range of 0.15−0.35, and 2 drugs with predicted probabilistic values <0.15. We further mapped the 16 drugs into LiverTox to extract likelihood scores defined based on consensus causality assessment of DILI cases by domain experts.[48] There were 5 drugs that overlapped with the LiverTox drug list. Of these 5 drugs, lopinavir had a predicted probabilistic value of 0.942. DeepDILI predicted three drugs, including elbasvir, darunavir, and dexamethasone, with probabilistic values of more than 0.35−0.95. Fingolimod was classified as category E* in LiverTox, meaning that the drug is suspected to be capable of causing liver injury or idiosyncratic acute liver injury, but there have been no convincing cases in the medical literature. DeepDILI predicted fingolimod with probabilistic values <0.15.

DeepDILI Model Could Be Used As a Screening Tool for Identifying the Potential DILI Concerns. The distribution of predicted probabilistic value of compounds in the Drugbank based on Mold2-DeepDILI is illustrated in Supplementary Figure S1. Of the 10,026 compounds, DeepDILI predicted 867 compounds (i.e., 867/10026 = 8.65%) with a probabilistic value more than 0.95, indicating the high DILI concern. Furthermore, 3763, 2486, 1655, and 1255 were

Table 4. Prediction Results of DeepDILI Model on the Repurposing Candidates for COVID-19[a]

Repurposing candidates	Chemical structure	Usage for COVID 19	DeepDILI prediction (probability)	LiverTox Category
Lopinavir		An HIV-1 protease inhibitor used in combination with ritonavir to treat human immunodeficiency virus (HIV) infection.	0.942	Category C https://www.ncbi.nlm.nih.gov/books/NBK548301/
Favipiravir		An antiviral used to manage influenza, and that has the potential to target other viral infections.	0.942	
TMC-310911		TMC-310911 (also known as ASC-09) is a novel investigational protease inhibitor (PI) that is structurally similar to the currently available darunavir. It is being investigated for use in HIV-1 infections.	0.914	
Triazavirin		An antiviral used to treat several strains of influenza.	0.89	
Metenkefalin		An investigational endogenous opioid being studied for the treatment of COVID-19.	0.887	
Umifenovir		A dual direct-acting antiviral/host-targeting agent used in the treatment and prophylaxis of influenza and other respiratory viruses.	0.886	

https://dx.doi.org/10.1021/acs.chemrestox.0c00374
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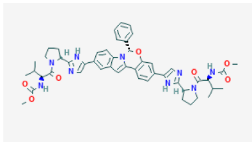
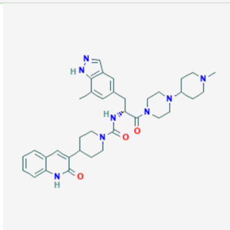
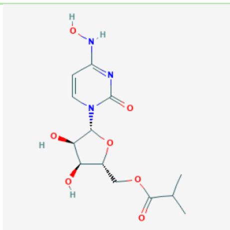
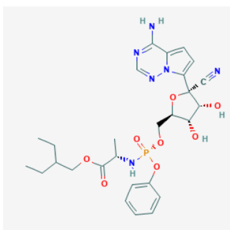
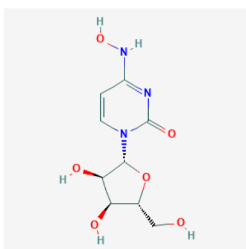
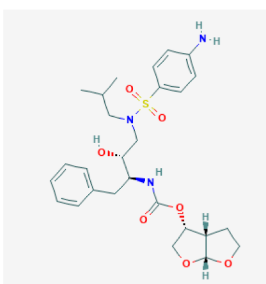
Table 4. continued

Repurposing candidates	Chemical structure	Usage for COVID 19	DeepDILI prediction (probability)	LiverTox Category
Elbasvir		An antiviral and NS5A inhibitor used to treat hepatitis C infections.	0.783	Category E* https://www.ncbi.nlm.nih.gov/books/NBK548933/
Vazegepant		Vazegepant is an antagonist of the calcitonin gene-related peptide (CGRP) receptor currently in phase 3 trials in an intranasal formulation for the treatment of migraine.	0.749	
EIDD-2801		An orally bioavailable isopropylester cytidine analog being investigated to treat COVID-19.	0.701	
Remdesivir		A nucleoside analog used to treat RNA virus infections including COVID-19.	0.673	
N4-Hydroxycytidine		A cytidine analog being investigated to treat COVID-19.	0.663	
Darunavir		A HIV protease inhibitor used in the treatment of human immunodeficiency virus (HIV) infection in patients with history of prior antiretroviral therapies.	0.658	Category D https://www.ncbi.nlm.nih.gov/books/NBK547994/

https://dx.doi.org/10.1021/acs.chemrestox.0c00374
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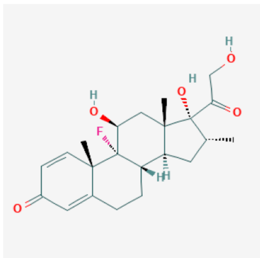
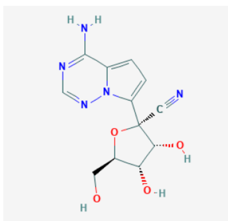
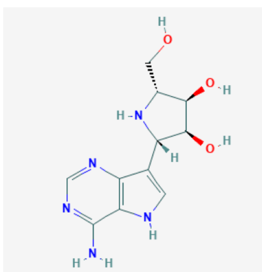
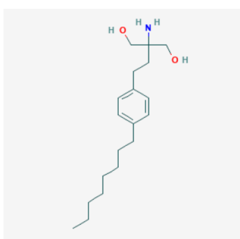
Table 4. continued

Repurposing candidates	Chemical structure	Usage for COVID 19	DeepDILI prediction (probability)	LiverTox Category
Dexamethasone		A glucocorticoid available in various modes of administration that is used for the treatment of various inflammatory conditions, including bronchial asthma, as well as endocrine and rheumatic disorders.	0.499	https://www.ncbi.nlm.nih.gov/books/NBK548400/
GS-441524		An experimental nucleoside analog antiviral investigated for its use in the treatment of Ebola and SARS-CoV-2 infections.	0.195	
Galidesivir		Galidesivir is an adenosine analogue that has been investigated for use against Zaire Ebolavirus. In animal studies, galidesivir was effective in increasing the survival rates from infections caused by various.	0.133	
Fingolimod		A sphingosine 1-phosphate receptor modulator used to treat patients with the relapsing-remitting form of multiple sclerosis (MS) and studied to manage lung complications of COVID-19.	0.119	Category E* https://www.ncbi.nlm.nih.gov/books/NBK548369/

[a]DILI likelihood scores were curated from the LiverTox database: https://www.ncbi.nlm.nih.gov/books/NBK548392/.

predicted as likely high DILI concerns (i.e., predicted probabilistic values in a range: 0.75−0.95), uncertain (i.e., predicted probabilistic values in a range: 0.35−0.75), likely less-DILI (i.e., predicted probabilistic values in a range: 0.15−0.35), and no-DILI concern (i.e., predicted probabilistic values <0.15), respectively (see Supplementary Table S1).

■ DISCUSSION

DILI is a significant safety concern in drug development, leading to clinical trial failures and drug withdrawals from the market, often attributable to complicated intrinsic and idiosyncratic mechanisms.[2] Tremendous efforts have been made in DILI risk prediction in the preclinical setting. However, prediction performance is suboptimal, and more efforts are still needed. In this study, we developed a DeepDILI model by integrating a model-level representation from conventional ML classifiers into a deep learning framework. The DeepDILI model aims to provide an explainable model and address the critical questions in the DILI field, such as using accumulative DILI knowledge from approved drugs to infer DILI potential for newly approved drugs. Encouragingly, the proposed DeepDILI model outperformed the conventional ML classifiers and the state-of-the-art ensemble methods, demonstrating great potential toward real-world application for early evaluation of DILI potential. To further facilitate our proposed DeepDILI model's real-world application, we used the whole DILIst data set, including 1002 drugs, to develop a full DeepDILI model product based on Mold2 descriptors (Supplementary Figure S2). The full DeepDILI model product can be accessed through https://github.com/TingLi2016/DeepDILI.

There have been a few attempts at deep learning approaches for DILI prediction but with limited success.[28] The sample size is always an obstacle in fully taking advantage of deep learning

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algorithms. Considering hundreds and millions of NN parameters, a limited number of samples could not make deep learning models master the information embedded in the data sets. Furthermore, DILI is a multifactorial end point, involving different underlying biological mechanisms. As a result, we have replaced molecule-level representation with model-level representation from the individual ML classifier to maximally extract DILI-related information from the data. Based on the current data set's performance results, the DeepDILI models could effectively integrate the information extracted from base classifiers and provide better and more balanced prediction than the molecule-level representation-based DNNs.

Two significant benefits of ensemble models are better prediction performance and more stable model results. We observed that individual base classifiers' performance fluctuated greatly, indicating the individual model captured different data information and correlated with the end point. The DeepDILI is a stacked model that harnesses a range of base classifiers' capabilities and makes predictions with better performance than any single classifier. Therefore, the diversity of base classifiers is essential to cover the data variance, enabling the meta-classifier to yield better performance. In our proposed DeepDILI model, we only selected the base classifiers that had a MCC in the 5% percentile to 95% percentile to improve the model robustness. Furthermore, unlike the conventional ensemble approaches (e.g., majority voting method and average probability method), we employed NN to maximally extract the useful information from base classifiers through an optimization theory for enhancing model performance.

Numeric molecular representations have been developed to be either fed into the ML model development or similarity comparison.[41] Besides a variety of descriptors and molecular fingerprints, some molecular representation based on artificial NNs[49,50] or natural language processing-inspired techniques such as Mol2vec[41] also emerged and showed some promising performances to represent chemical information and enhance the model performance. It is worth investigating the impact of different molecular representations on DeepDILI performance. In this study, we compared the performance of DeepDILI models with the three different molecular descriptors, including Mold2, Mol2vec, and MACCS. We found that the Mold2-DeepDILI model outperformed the other two Mol2vec-DeepDILI model and MACCS-DeepDILI. However, the conclusions were drawn only based on the current data sets and model framework. More investigation of molecular descriptors' impact on model performance should be further investigated with diverse biological end points under different ML algorithms.

One of the critical aspects of implementing a computational model for real-world application is model explainability, where explainability refers to the ability of parameters, often hidden in deep nets, to justify the results. In our proposed DeepDILI model, a sequential feature selection strategy was developed to generate a more explainable model with important feature subsets. These informative descriptors are very helpful in associating human DILI risk with the specific chemical-physical properties. We obtained a list of 21 descriptors each of which accounted for more than 3% of model contributions. Several descriptors have been reported to play essential roles in hepatotoxicity. For example, descriptors such as '*classic aromatic antiepileptic drugs with aromatic bonds*' and '*number of group secondary (aromatic) amides*' are limited in their clinical use mainly because of idiosyncratic hepatotoxicity.[51] Furthermore, it was reported that several carboxylic acid-containing drugs, such as ibufenac and benoxaprofen, have been withdrawn postmarketing due to overt liver toxicity.[52]

A defined applicability domain is of great importance to position the developed DeepDILI model as "fit-for-purpose" in real-world applications. The model performance by the therapeutic class demonstrated that the developed DeepDILI model had a superior prediction performance (i.e., MCC = 0.581) for the 'A, alimentary tract and metabolism' category. Because of the diverse chemical class within the alimentary tract and metabolism category and the complicated underlying biology mechanisms, it is a challenging task to identify the potential risk for clinical hepatotoxicity.

It should be noted that some additional investigations may be merited in order to improve the DeepDILI model even further. First, we arbitrarily choose the base classifiers based on their performance in the range of 5−95%, on the basis that the high-performance models may have overfitting problems and the low-performance models may have underfitting questions. Second, base classifier diversity can be further improved by using multiple chemical representations, such as ECFP4,[53] MACCS,[42] etc. Several studies have indicated that the ensemble models can be enhanced by combining base classifiers built from multiple chemical representations.[24,35,54] Finally, in our current study, we only use 10 neurons in the NN hidden layer and in tuning the essential parameters, including activation and optimizers. Other parameters, such as the number of neurons and the number of hidden layers, were selected by experience. Because of this, the predictability of the developed model may be improved by further fine-tuning.

DILI is a multifactorial end point that current preclinical animal models may miss. Our developed DeepDILI model has great potential to overcome this challenge, which not only can help to reduce drug attrition for the pharmaceutical industry but also allow patients to reduce the chance of exposure to DILI risk. Furthermore, we propose the DeepDILI model as a basis to further explore novel deep learning frameworks for augmenting DILI prediction. We hope our paper will trigger interest in further exploring advanced artificial intelligence approaches to eliminate DILI potential at the earliest stage of drug development and in repurposed drug candidates.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.chemrestox.0c00374.

Figure S1: The distribution of predicted probabilistic values of compounds in the Drugbank based on Mold2-DeepDILI (PDF)

Figure S2: Workflow of the DeepDILI model with Mold2 descriptors based on the whole data set (i.e., the training and test set) (PDF)

Table S1: Data information and prediction results on the training, test, three external test sets and DrugBank (XLSX)

Table S2: Hyperparameter optimization of the five conventional ML methods based on the seven performance metrices on the training set 2 (XLSX)

Table S3: Ranked descriptors of Mold2, MACCS, and Mol2vec-based on the model contribution (XLSX)

Table S4: Model performance of base classifiers (i.e., LR, KNN, SVM, RF, and XGBoost on the test set) with the whole training set (XLSX)

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Author Contributions

Z.L. and W.T. conceived and designed the study. T.L. and Z.L. performed data analysis. T.L., Z.L., and W.T. wrote the manuscript. R.R., S.T., Z.L., and W.T. revised the manuscript. All authors read and approved the final manuscript.

Notes

The views presented in this article do not necessarily reflect current or future opinions or policies of the US Food and Drug Administration. Any mention of commercial products is for clarification and not intended as an endorsement.

The authors declare the following competing financial interest(s): R.R. is Cofounder and Codirector of ApconiX, an integrated toxicology and ion channel company that provides expert advice on nonclinical aspects of drug discovery and drug development to academia, industry, and not-for-profit organizations.

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